

Optimal Dispatch of Electric Vehicle Batteries between Battery Swapping Stations and Charging Stations

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Abstract— Along with the fast development of battery technologies and growing criticism on heavy air pollution caused by traditional automobiles, electric vehicles (EV) draw more attention in these years. The inconvenience of battery charging is the main barrier to the widespread use of EVs, and the EV battery-swapping technology is a promising method to assist overcoming this difficulty eventually because of its flexibility. In this paper, a battery schedule framework is studied to dispatch batteries between battery charging stations (BCS) and battery swapping stations (BSS) efficiently. A two-direction battery dispatch model to reduce the transportation cost is established and solved by PSO method. Moreover, considering the serving ability limitations, the K-means clustering algorithm is utilized to pre-partition the BCS and BSS to make the battery dispatch more efficient and effective. The proposed methods is finally verified by an urban battery logistics case.

Index Terms— electric vehicle, battery charging station, battery swapping station, optimal dispatch.

I. INTRODUCTION,

RAPID development of electric vehicles (EV) takes along with new opportunities as well as challenges to the traffic system, power system and automobile industry. Widespread use of EVs will reduce the vehicles' consumption of fossil oil and contribute to environmental conservation [1]. However, the fast growing of EV charging demand also brings negative impacts to the power system operation [2]. Because of the uncertainty nature of EV charging behavior, the energy losses may be much higher in a distribution system with a high penetration level of EVs [3]. The power quality and economic efficiency of the power grid may also decrease because of the random charging of a large amount of EVs [4].

Another main problem significantly limits the popularization of EVs is the inconvenience of EV charging. Long charging time and lack of charging facilities reduce consumers' willingness of buying EVs remarkably [5]. EVs usually take hours of time for charging, which is a great concern for most EV consumers [6]. Battery swapping

method is very promising to overcome those difficulties [7]. Besides, battery swapping stations (BSS) are potential to provide service to the power grid. For example, BSS have a better performance on smoothing the load profile than fast charging stations (FCS) because of the flexible battery charging characteristics [8].

Research on the optimal operation of BSS has begun in recent years. A charging power dispatch model is established in [9] to reduce fluctuations and peak-valley difference of load demand in the distribution system. A 2-stage optimal charging model with the objectives of charging cost minimization and charging power smoothing is studied in [10] considering the bus running operation and battery discharging characteristics. Unit commitment and economic dispatch problems of smart grid with BSS are investigated in [11] and [12].

Although some progress is obtained on the research of BSS, the in-depth study of BSS operation is rather underdeveloped [9-12]. Most existing research treats BSS as a special kind of load, and ignores its differences from common loads. Actually, because the battery charging process is separated from vehicles in the battery swapping mode, the battery swapping charging network is usually consisted by BSS and battery charging stations (BCS) independently. In this mode, the battery logistic between BSS and BCS becomes an urgent problem to ensure the economic and safety operation of the battery swapping charging network. However, the study on the battery transportation between BCS and BSS is quiet insufficient. In this work, the battery swapping charging network is studied first, and the optimal dispatch of batteries between BCS and BSS is investigated and solved by the Particle Swarm Optimization (PSO) method. Considering the fast development and extensive usage of EVs in the future, the pre-partition of BCS and BSS according to the geographic location is quiet necessarily and solved by a novel K-means clustering algorithm in this paper. The feasibility and efficiency of the

developed dispatch model is demonstrated with an urban battery logistics case. The results show that compared with random transportation, optimal dispatch of EV batteries is feasible and reduces the battery transportation cost efficiently, which will contribute to the development of battery swapping technologies and the wide application of EVs in the future.

II. ASSUMPTIONS AND MODELING

In this section, the structure of the battery swapping charging network and its operation mechanism are firstly described. Then the basic model and extended model of battery dispatch is established and analyzed.

A. Battery Swapping Charging Network

As shown in Fig. 1, in the proposed battery swapping charging mode, EVs exchange their exhausted batteries with fully charged batteries in the BSS, and the batteries exchanged from EVs in the BSS will be transferred to the BCS, where a large number of batteries could get charged uniformly. In addition, the batteries already charged are transported to BSS for the usage of other EVs. The remarkable advantage of this mode is that the BSS only exchange batteries thus does not occupy large space, and the BCS could charge batteries in a more organized way, which will greatly improve the charging efficiency and reduce the charging service cost.

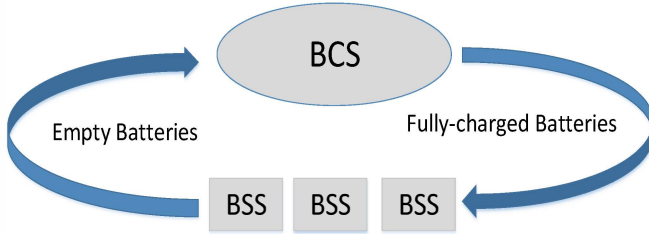


Fig 1. Battery swapping charging network structure

B. The Basic 2-Direction Battery Dispatch Model

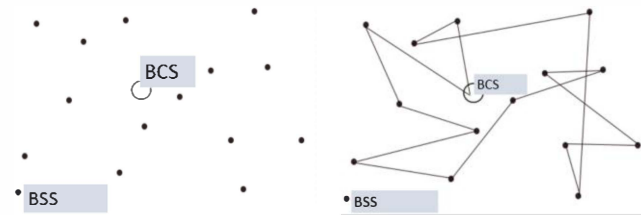


Fig 2. Basic battery dispatch model

Firstly the basic model of battery dispatch is described. The objective of the proposed model is to found the shortest pathes to delivery full-charged batteries from BCS to BSS and get the empty batteries back from the BSS to BCS. In this model the transportation node set are $V = \{v_i | i = 0, 1, 2, \dots, n-1\}$ and the path set are $E = \{(v_i, v_j) | v_i, v_j \in V, i \neq j\}$. The BSS location set are $C = \{v_1, v_2, \dots, v_{n-1}\}$ and the BCS is denoted as v_0 . $C_y = C \cdot d_{ij}$ is the transportation cost, where C_0 is the cost per km and d_{ij}

is the travel distance. $A = \{g_i | i = 1, 2, \dots, m\}$ is the set of delivery vehicles and the total number of which is m . The battery demand is q_i and G is the delivery vehicle capacity. The objective function and constraints are as follows:

Objective

$$\min Z = \sum_{i=1}^m \sum_{j=0}^{n-1} \sum_{k=1}^m C_{ij} X_{ijk} \quad (1)$$

Constraints

$$\sum_{k=1}^m \sum_{j=1}^{n-1} X_{0jk} \leq m \quad (2)$$

$$\sum_{k=1}^m \sum_{i=0}^{n-1} X_{ijk} = 1, \forall v_j \in C \quad (3)$$

$$\sum_{k=1}^m \sum_{j=0}^{n-1} X_{ijk} = 1, \forall v_i \in C \quad (4)$$

$$\sum_{i=1}^{n-1} X_{i0k} = 1, \forall k \in A \quad (5)$$

$$\sum_{i=1}^{n-1} X_{0ik} = 1, \forall k \in A \quad (6)$$

$$\sum_{k=1}^m \sum_{j=1}^{n-1} X_{0jk} - \sum_{k=1}^m \sum_{i=1}^{n-1} X_{i0k} = 0, \forall k \in A \quad (7)$$

$$\sum_{i=1}^{n-1} q_i \sum_{j=0}^{n-1} X_{ijk} \leq G, \forall k \in A \quad (8)$$

$$X_{ijk} = \begin{cases} 1 & \text{if the } k\text{th vehicle pass } i \text{ and } j \\ 0 & \text{otherwise} \end{cases} \quad \forall v_i, v_j \in V, \forall k \in A \quad (9)$$

where Equ. (1) is the objective function to minimize the delivery cost; Equ. (2) is to ensure the delivery vehicle number limitation; Equ. (3) and (4) means every BSS's battery is provided by only 1 vehicle; Equ. (5) and (6) means every vehicle only travel one time. Equ. (7) ensures that every vehicle could go back to the BCS; Equ. (8) limits the vehicle transportation capacity and the Equ. (9) is the integer constraint.

C. The Extended Battery Dispatch Model by K-means Pre-partition Method

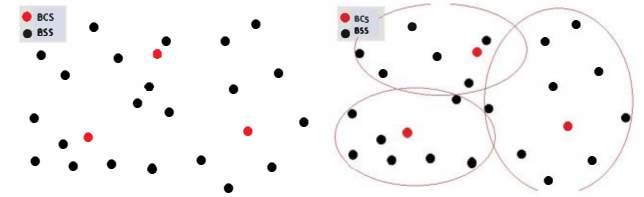


Fig 3. The pre-partition step of the extended battery dispatch model

With the development of electric vehicle and battery technologies, the wide application of EVs is possible in the future. The optimal dispatch of a battery swapping charging network with a high penetration of EVs in a large area is a great challenge for the system operators. In this situation, there are more difficulties for the battery dispatch like longer travel distance, higher delivery capacity and service ability limitations, et al.

A promising way to solve this problem is pre-partition the service areas, as shown in Fig. 3. After the pre-partition, the

battery dispatch problems could be solved separately, which will reduce the service burden and make the dispatch more efficient.

K-means clustering is popular for cluster analysis and partitions n observations into k clusters in which each observation belongs to the cluster with the nearest mean [13]. The K-means clustering could be applied into our problem as follows:

- 1) Set the c BCS locations as the initial centers;
- 2) In the k th iteration, calculate the within-cluster sum of squares (simply Euclidean distances here) of every BSS to c BCS, and partition the BSS to the nearest BCS cluster;
- 3) Update the center locations of the c BCS clusters according to least-squares estimator (arithmetic means, et. al);
- 4) Repeat step 2)-3) until the cluster assignment no longer change;
- 5) Output the cluster results.

The flowchart of the K-means clustering is shown in Fig. 4:

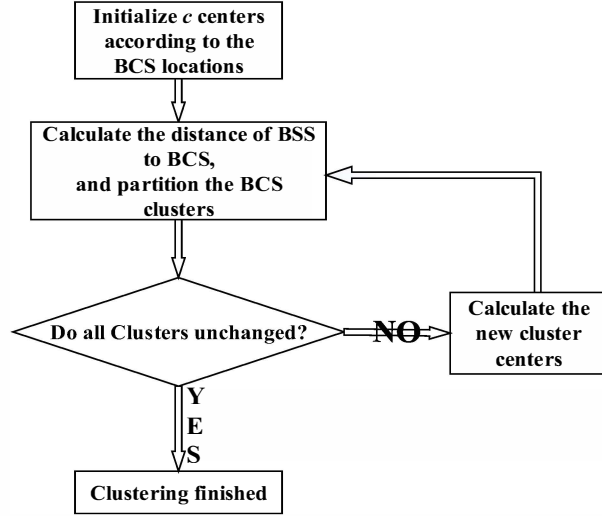


Fig 4. Flowchart of the BCS service K-means clustering

III. SOLVING STRATEGIES

In this paper, PSO algorithm employed to solve the optimization problem described before. First, the constraints are dealt with by the penalty function method. Second, the fitness function is formed by the objective function and penalty constraints together.

A. PSO Based Optimal EV Battery Dispatch

Step 1) Specify some parameters associated with PSO, such as population size n_p , initial solution x_i , et al;

Step 2) Partition the service area of BCS by the K-means clustering method;

Step 3) Calculate the objective function value of each cluster respectively;

Step 4) Initialize the particle swarm, velocity and acceleration;

Step 5) Generate the initial solution;

Step 6) Calculate the fitness value;

Step 7) Find the optimal values of individuals and the optimal solution of the particle swarm;

Step 8) Update the velocity, acceleration and individual values;

Step 9) Repeat Steps 3)-8) for N_c times.

Step 10) Select the best individual found in the above solving procedure as the optimal solution of the BSS battery dispatch.

The flowchart of the PSO procedure is shown in Fig. 5.

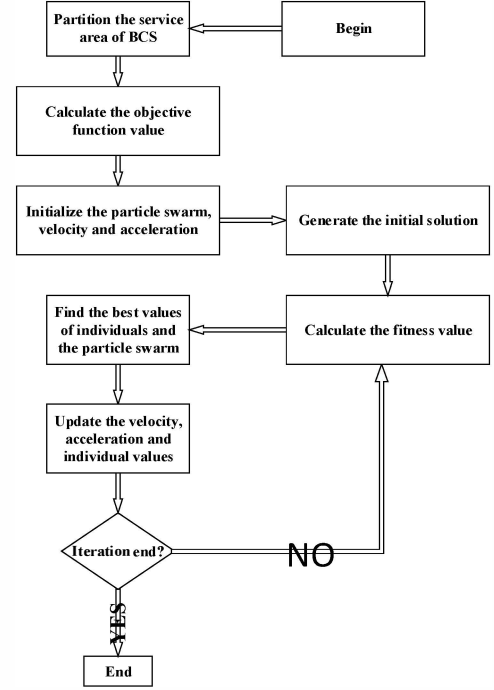


Fig 5. Flowchart of the PSO based optimal EV battery dispatch.

IV. CASE STUDIES

A. The Basic 2-Direction Battery Dispatch Model Case

Firstly, a simple case of a 100 sq.km. area with 1 BCS and 7 BSS is studied. The BCS and BSS location is generated randomly. The transportation cost parameter $C_1=C_2=0.5$ dollar, and the PSO group size is 200. Fig. 6 shows the random travel path to deliver all the batteries from BCS to BSS. The total distance is 62.3024 km.

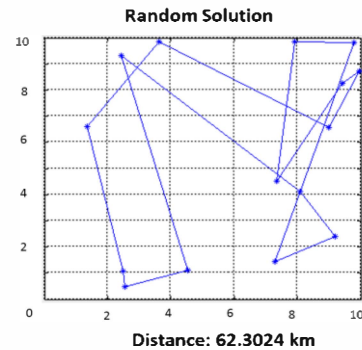


Fig 6. The random travel path of the basic battery dispatch model.

In Comparison, Fig. 7 shows the optimal travel path of the basic model. It is clearly demonstrated that after optimization, the travel distance is greatly reduced from 62.3024 km to 32.7437 km. Therefore, the travel cost will be saved a lot.

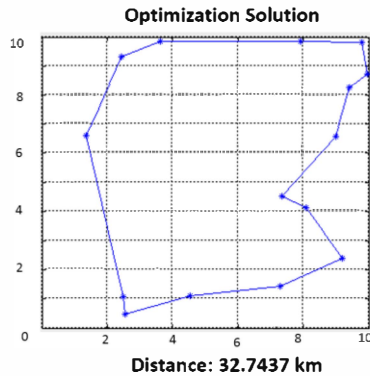


Fig 6. The optimal travel path of the basic battery dispatch model.

Fig. 7 shows the performance of the proposed PSO solving strategy. The objective function value becomes stable after 50 iterations and the optimal solution could be obtained.

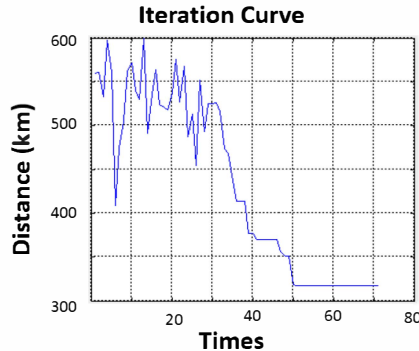


Fig 7. The iteration curve of the proposed PSO method.

B. The Extended Battery Dispatch Model Case with K-means Clustering Method

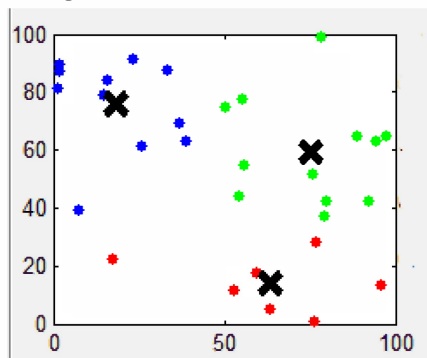


Fig 8. The BSS/BCS partition results in a larger area.

It can be concluded that the basic model is very effective in the small area case. However, when the service area is large enough with high penetration of EVs in the future (as shown in Fig. 8), the pre-partition is essential to solve the high-dimension problem separately. Tab. I shows the geographic locations in the extended model. There are 3 BCS and 27

BSS in this model. Fig 8. shows the partition results (Blue, red and green BSS means different clusters) in an area of 10000 sq.km.

Tab. I The BSS/BCS geographic location in the extended model

No.	X	Y	No.	X	Y
1	97.43	65.07	16	59.12	17.47
2	52.85	11.29	17	25.69	61.38
3	75.88	51.62	18	1.98	89.79
4	50.28	75.07	19	54.41	44.1
5	55.67	54.74	20	63.25	5.14
6	78.49	98.9	21	55.41	77.76
7	79.33	36.69	22	23.12	91.28
8	1.89	87.43	23	14.56	79.21
9	33.02	87.59	24	94.41	62.81
10	36.77	69.51	25	17.13	22.35
11	79.72	42.43	26	76.89	28.08
12	88.51	64.62	27	15.91	84.58
13	76.14	1.05	28	38.82	62.81
14	7.03	39.09	29	95.62	13.43
15	92.41	42.34	30	1.11	81.53

After partition, the optimal battery dispatch model could be solved separately. Fig. 9 shows the random and optimal battery delivery path before partition. It can be observed that the shortest battery delivery path distance is 645.37 km before partition. Although it is much better result than the random solution of 1287.22 km, the distance is still a great burden for battery delivery tasks.

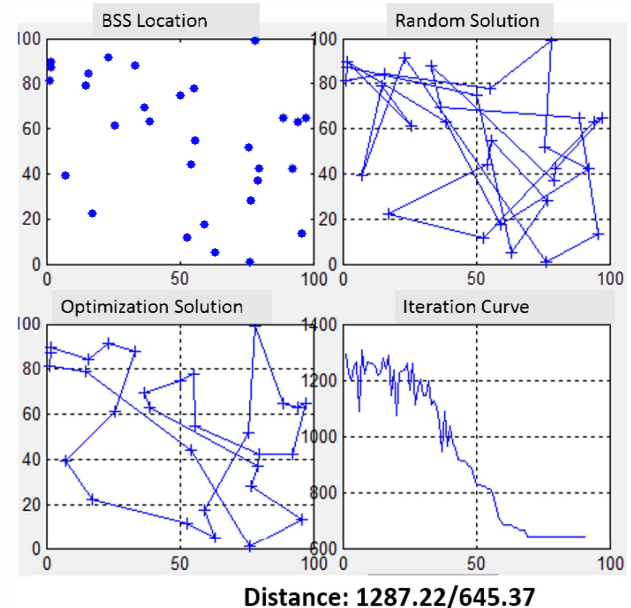


Fig 9. The random path, optimal path, and iteration curve of the extended model before partition.

Fig 10 shows the simulation results after partition. It clearly shows that the total battery delivery path distance is

greatly reduce from 645.37 km to 524.61 km, thus the BCS and BSS service ability is improved to promote to promote the wide application of EVs in the future.

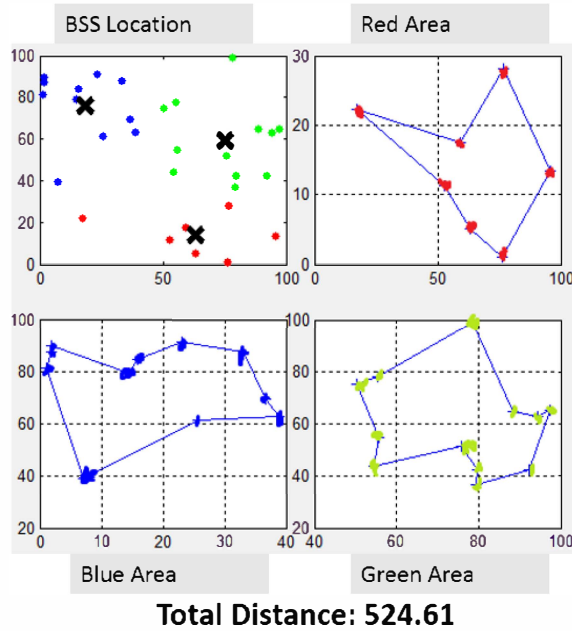


Fig 10. The optimal path of the extended model after partition.

V. CONCLUSION

Rapid deployment of EVs imposes significant challenges with respects to secure and economical operation of power systems. The planning of EV battery charging/swapping stations is quiet important to support the application of Evs. Currently, the research related to the optimal dispatch of batteries between BCS and BSS is still in the early stage. Hence, this paper has established an original battery optimal dispatch model to take full advantages of the battery swapping mode vehicles, thus largely contributing to optimal operation of the battery swapping charging network in power systems. Case studies have indicated that the optimal dispatch model based on PSO algorithm and K-means partition method can efficiently dispatch batteries and reduce the traveling distance, thus largely reducing the operation costs of the existing battery swapping charging networks.

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REFERENCES

- [1] W. Su, H. Tahimi-Eichi, W. Zeng, and M. Chow, "A survey on the electrification of transportation in a smart grid environment," *IEEE Trans. Ind. Informat.*, vol. 8, no. 1, Feb. 2012.
- [2] Z. Darabi and M. Fedowsi, "An event-based simulation framework to examine the response of power grid to the charging demand of plug-in hybrid electric vehicles (PHEVs)," *IEEE Trans. Ind. Informat.*, vol. 10, no. 1, pp. 313-322, Feb. 2012.
- [3] L. P. Fernández, T. G. S. Román, R. Cossent, C. M. Domingo, and P. Frias, "Assessment of the impact of plug-in electric vehicles on distribution networks," *IEEE Trans. Power Syst.*, vol. 26, no. 1, pp. 206-213, Feb. 2011.
- [4] K. Clement-Nyns, E. Haesen, and J. Driesen, "The impact of charging plug-in hybrid electric vehicles on a residential distribution grid," *IEEE Trans. Power Syst.*, vol. 26, no. 1, pp. 371-380, Feb. 2010.
- [5] G. Wang, Z. Xu, F. Wen, and K. P. Wong, "Traffic-constrained multiobjective planning of electric-vehicle charging stations," *IEEE Trans. Power Del.*, vol. 28, no. 4, pp. 2363-2372, Oct. 2013.
- [6] G. Wang, J. H. Zhao, F. Wen, Y. Xue, and G. Ledwich, "Dispatch strategy of PHEVs to mitigate selected patterns of seasonally varying outputs from renewable generation," *IEEE Trans. Smart Grid.*, vol. 6, no. 2, pp. 627-639, Mar. 2015.
- [7] Q. Dai, T. Cai, S. Duan, and F. Zhao, "Stochastic modeling and forecasting of load demand for electric bus battery-swap station," *IEEE Trans. Power Del.*, vol. 29, no. 4, pp. 1909-1917, Aug. 2014.
- [8] Y. Zheng, Z. Y. Dong, Y. Xu, K. Meng, J. H. Zhao, and J. Qiu, "Electric vehicle battery charging/swap stations in distribution systems: comparison study and optimal planning," *IEEE Trans. Power. Syst.*, vol. 29, no. 4, pp. 1909-1917, Aug. 2014.
- [9] W. Tian, J. He, Y. Jiang, et al, "Multi-objective optimization of charging dispatching for electric vehicle battery swapping," *Power Syst. Tech.*, vol. 36, no. 11, pp. 25-29, 2012. (in Chinese)
- [10] Y. Yang, Z. Hu, and Y. Song, "Research on optimal operation of battery swapping and charging station for electric buses," *Proc. of the CSEE*, vol. 32, no. 31, pp. 35-42, 2012. (in Chinese)
- [11] S. Zhang, Z. Hu, Y. Song, et al, "Research on unit commitment considering interaction between battery swapping station and power grid," *Proc. of the CSEE*, vol. 32, no. 10, pp. 49-55, 2012. (in Chinese)
- [12] D. Zhang, Y. Jiang, W. Zhang, et al, "Economic operation of electric vehicle battery swapping station based on genetic algorithms," *Power Syst. Tech.*, vol. 37, no. 8, pp. 2102-2107, 2013. (in Chinese).
- [13] S. P. Lloyd, "Least squares quantization in PCM," *IEEE Transactions on Information Theory* vol. 28, no. 2, pp. 129-137, 1982.